

Truly-Differential Drive of TFLN TWE-MZM by Linear SiGe Driver in a Codesigned Hybrid Integrated Assembly

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Abstract: We present a hybrid assembly of a SiGe driver and TFLN modulator, featuring a novel differential driving scheme achieving low $V_{\pi L} = 1.1$ Vcm. We demonstrate 140 GBd PAM-8 modulation with 1.4 pJ/bit energy consumption.

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1. Introduction

The ever-growing internet traffic sets demanding targets on the optical infrastructure. In particular, optical transmitters are required to show extremely high bandwidth (BW) and maintain excellent power performance. Thin-film Lithium Niobate (TFLN) on Silicon based Mach-Zehnder modulators have been shown to deliver electro-optic (EO) BWs beyond 100 GHz, but, due to the low efficiency of their Pockels electro-optic effect, these bandwidths can only be achieved in conjunction with a switching voltage larger than 3 V [1,2]. This value is not only well beyond what can be delivered by CMOS ASICs, but at this speed becomes challenging even for fast IC technologies such as InP and SiGe. For this, differential driving schemes have been commonly used to drive MZMs due to their higher available swing, and further benefits such as superior resistance to electromagnetic and power supply noise, higher common-mode rejection, and lower static power consumption. Although differential driving schemes are commonly employed for modulator platforms such as InP or Si, Pockels-based planar configurations such as TFLN make this harder due to the directionality of the linear effect with the electric field. Conventional differential configurations like GSGSG or GSSG are not inherently compatible with the three-conductor travelling-wave electrode (TWE) used in TFLN-MZMs. Efforts to adapt these schemes often result in the loss of the push-pull effect, thereby forfeiting the voltage-swing benefits associated with differential driving [3]. Recently, some differentially driven modulators have been presented which attempt to solve the issue by modifying the structure of the TWE either in between the electrodes or by splitting and rearranging the TWE itself in a GSGSG or GSSG fashion [4,5]. However, these approaches come at the cost of a loss of performance and/or significantly increased PIC and setup complexity. Instead of acting on the modulator alone, the solution we present here is based on the codesign of the MZM and its driver IC.

2. Differential MZM and Driver IC

The schematic of the codesigned components and their interconnection is shown in Figure 1(a). The driver IC is modified to deliver a custom $\overline{SSSS}$ output signal with no ground planes in between the signal electrodes. Our approach enables a full differential drive leveraging the push-pull effect and effectively doubles the modulation efficiency of the TFLN-MZM with minimal changes.

The C-band I/Q modulator is a custom design fabricated in a commercial foundry multi-project wafer run with 350 nm of Lithium Niobate on a Silicon substrate. The TWE is 7.5 mm long with characteristic impedance of 50 Ω and has periodic capacitive loading and integrated NiCr termination resistors. The 4-pad electrical input of each channel is designed to allow for the direct bonding to the differential-output driver IC with the two inner pads connected to the same central electrode. Besides this input modification, the inner part of the TWEs is identical to the one of a modulator to be driven in a standard GSG fashion and can be characterized stand-alone in such a way, which means a compact and simple design (no ground planes are required in between the signal lines) with straight-forward operation. The V_{π} of the modulator is measured to be 3.26 V, which translates to a very low $V_{\pi L} = 1.1$ Vcm referenced to the single-ended signal thanks to the push-pull differential drive. The 3-dB EO bandwidth of the modulator, shown in Figure 1(b), exceeds 67 GHz, limit of the measurement equipment at the time of characterization.

The dual-channel I/Q driver IC is implemented in a commercial 130-nm SiGe BiCMOS technology with 350 GHz f_T and 450 GHz f_{max} . Each channel is independently controllable and contains two gain stages for a total of 20 dB gain and two emitter follower (EF) stages for level shifting and frequency response peaking. The first gain stage is a variable gain amplifier (VGA) with up to 15 dB gain, set via an analog control voltage accessible through a wire-bonded pad. For optimum power efficiency, an open-collector (OC) stage is employed at the driver output with a fixed gain of 5 dB for a 50 Ω load. It is split into two parallel output stages, each driving a pair of differential pads of the $\overline{SSSS}$ configuration. The signal is split between the two EF stages, a symmetrical layout ensures good matching even at high frequencies. The DC and signal path extend into the TWEs, terminated by monolithically integrated resistors connected to the 5 V supply voltage via wire-bonds. This approach requires close integration with short interconnects to avoid reflections at the unmatched IC outputs. The driver IC consumes 605 mW per channel: 39 mA from 4.4 V (VGA and EF1), 45 mA from 3.3 V (EF2), and 57 mA from 5 V supply (OC). The differential signal doubles the available voltage between the electrodes compared with single-ended drive, which increases the electro-optical efficiency of the assembly. In addition, it reduces nonlinearities by eliminating even harmonics enabling higher modulation orders such as PAM-8 and QAM-64. When terminated on 50 Ω , the maximum available differential voltage swing is simulated to be 2.8 V for each of the two open-collector output stage pairs with 3-dB bandwidth > 90 GHz.

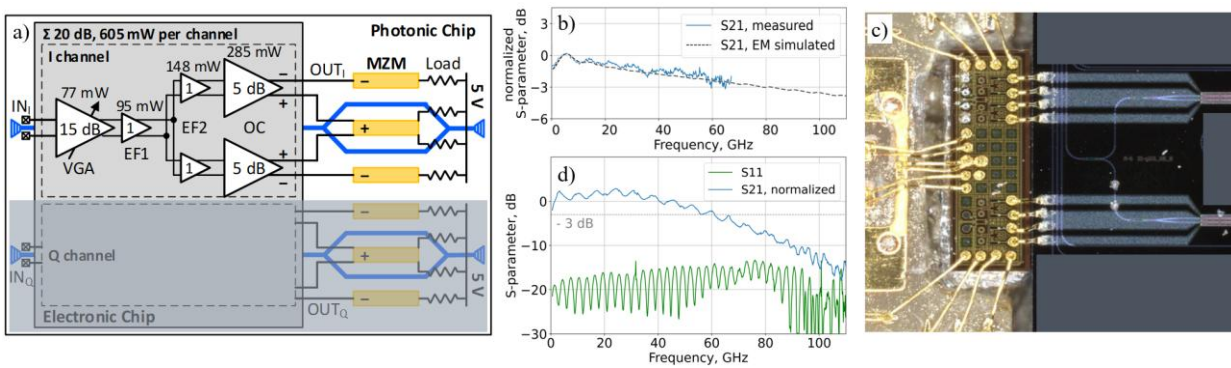


Fig. 1. a) block diagram of the fully differential electronic and photonic chip. Only the I-channel is characterized in this work, b) EO frequency response of the TFLN modulator, c) microscope detail of the IC and interconnects to the MZM, and d) frequency response of the differential IC-MZM assembly

Both the driver IC and the MZM are designed to support I/Q operation, but only a single channel could be characterized in this work due to lack of availability of a dual-channel input probe for the IC. The components are placed on an aluminum block and wire-bonded to a PCB for DC biasing. The PCB has a cut-out to place the driver close to the TFLN modulator while keeping them at similar height for minimized bond-wire length, as shown in Figure 1(c). The small-signal response of the assembly is obtained by directly probing the differential driver inputs of channel I with a 110-GHz GSSG probe and connecting the two signal inputs to two ports of a vector network analyzer (VNA). The modulator output is captured with a 110-GHz photodiode connected to a third port of the VNA. Fig. 1(d) shows the measured input reflection and electro-optic transmission of the assembly with good differential input matching up to 110 GHz and a 3-dB bandwidth of about 60 GHz with a mid-gain setting for the IC. The ripples observed in the joint S21 are due to the fabricated terminations exhibiting a resistance value lower than targeted (35 Ω instead of 50 Ω). Considering the components' BWs, a significant penalty coming from the wire-bonded assembly can be observed. This will need to be addressed with more sophisticated assembly methodologies in future.

3. Experiment

We perform an intensity-modulation/detection experiment to test the large signal performance. A standard 40 kHz linewidth external cavity laser is used as the light source and amplified to 23 dBm by an EDFA before being sent to the IC-MZM assembly. This high input power is necessary to compensate for the large coupling losses due to mode size mismatch with the available fibers at the time of the assembly, estimated to be > 15 dB. The MZM is biased at the quadrature point and the driver IC is operated at max gain. One channel of the integrated driver is driven by a 256 GSa/s DAC that is interfaced by a GSSG probe. The test signal is an RRC-shaped ($\beta=0.01$) PAM-4/8 symbol sequence that is digitally pre-emphasized to compensate for the combined linear frequency response of the transmitter: i.e. DAC, cables, driver, MZM. The modulated output of the MZM is amplified by a

second EDFA to around 6 dBm before detection on a single 100 GHz bandwidth photodiode and acquisition by a 256 GSa/s oscilloscope for offline processing. The digitized PAM signals are then processed via standard digital signal processing at 2 samples per symbol including a 601-tap least-means-squares (LMS) equalizer embedded with a digital phase locked loop. The normalized generalized mutual information (NGMI) [6] of the received signal is calculated and plotted in Figure 2 as the symbol rate is varied from 120-180 GBd for both PAM-4 and PAM-8. The NGMI threshold is set by a forward error correction (FEC) scheme that concatenates a soft-decision spatially-coupled low-density parity-check (SC-LDPC) inner code (rate 5/6) and a hard-decision Bose–Chaudhuri–Hocquenghem (BCH) outer code (rate 0.9915), leading to a concatenated code rate of 0.8262 with an NGMI threshold of 0.8714 [7]. The maximum error-free rate with this FEC is achieved at 140 GBd PAM-8, leading to a single dimension net data rate of 347 Gb/s. Considering the gross data rate, the energy consumed per bit of the IC-MZM assembly is 1.4 pJ/bit, which is significantly lower than previously reported in alternative hybrid assemblies [3]. The NGMI value for PAM-4 at 180 GBd is well above threshold and therefore the differential IC-MZM assembly is likely to support error free transmission at significantly higher baudrates.

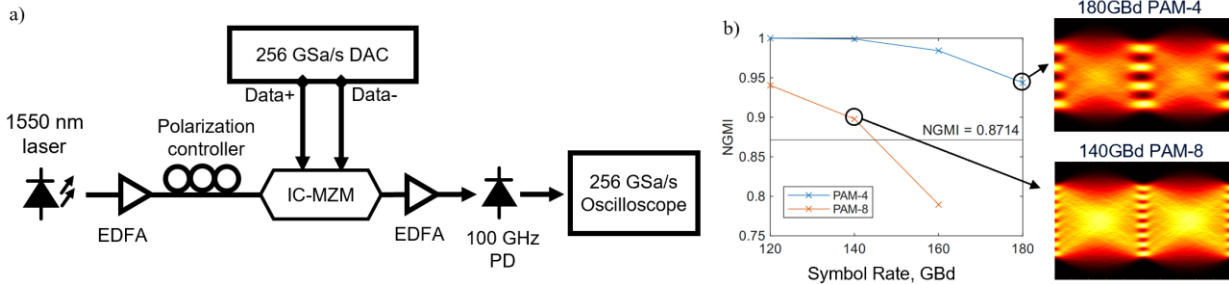


Fig. 2. a) schematic of the large signal measurement setup for a single driver and modulator channel, and b) NGMI of the modulated signal for PAM-4 and PAM-8 at different symbol rates.

4. Conclusion

We demonstrate a hybrid integrated assembly with codesigned SiGe driver IC and TFLN-MZM enabling truly differential push-pull drive voltage. A power efficiency of 1.4 pJ/bit can be achieved with 140 GBd PAM-8 transmission while considerable NGMI headroom exists with PAM-4 signals up to 180 GBd and beyond. This combination provides a path towards 400 Gbit/s per lane with commercially available semiconductor technologies and a simple wire-bond assembly. Using advanced SiGe BiCMOS and TFLN technology platforms for electronic and photonic chips, as well as advanced assembly techniques like 3D flip-chip integration could significantly boost performance towards higher data rates in the future.

5. References

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